DATA ITEM DESCRIPTION

Form Approved
OMB No. 0704-0188

1. TITLE	2. IDENTIFICATION NUMBER
Endurance Fuel Calculations (Non-Nuclear Ships)	DI-MISC-80852

3. DESCRIPTION / PURPOSE

3.1 Endurance fuel calculations provide overall ship fuel performance data and determine the necessary fuel tankage required to meet specified ship endurance requirements.

4. APPROVAL DATE (YYMMDD)	5. OFFICE OF PRIMARY RESPONSIBILITY (OPR)	6a. DTIC APPLICABLE	6b. GIDEP APPLICABLE
890614	N-NAVSEA 56X11		

7. APPLICATION / INTERRELATIONSHIP

7.1 This DID contains the format and content preparation instructions for the data product generated by the specific and discrete task requirement as delineated in the contract.
7.2 These data apply to conventionally powered steam, diesel, gas turbines (or combination thereof) ships or crafts.
7.3 This DID supersedes UDI-E-23107B.

8. APPROVAL LIMITATION	9a. APPLICABLE FORMS	9b. AMSC NUMBER
		N4748

10. PREPARATION INSTRUCTIONS

10.1 Content. The endurance fuel calculations shall include all data indicated on Figure 1.. In the case of diesel or gas turbine powered ships, fuel performance curves for individual components using their own fuel system (i.e., main propulsion diesel or gas turbines engines, ship service generators, auxiliary boilers, etc.) shall be included in the endurance fuel calculations.

10.2 Format. The endurance fuel calculations shall be presented in a method and format similar to Figure 1.

11. DISTRIBUTION STATEMENT

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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DESIGN _____
PREPARED BY _____
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EXAMPLES

	STEAM	DIESEL OR GAS TURBINE
(1) Endurance Required, Nautical Miles	3,000	1,200
(2) Endurance Speed, Knots	15	6
(3) Full Load Displacement, tons	3,000	400
(4) Rated Full Power, SHP	50,000	700
(5) Design Endurance Power @(2)&(3), SHP	3,000	150
(6) Average Endurance Power, SHP: (5) x 1.10	3,000 x 1.10 = 3,300	150 x 1.10 = 165
(7) Ratio, Avg. End SHP/rated F.P. SHP: (6)/(4)	0.066	0.24
(8) Average Endurance BHP: (6)/Transmission Efficiency	---	165/0.95 = 174
(9) 24-Hour Average Electric Load, kW	500	30
(10) Calculated Propulsion Fuel Rate @(8), lb/BHP/hr	---	0.479
(11) Calc. Prop. Fuel Consumption, lb/hr: (10) x (8)	---	0.479 x 174 = 83.4
(12) Calc. S.S. Gen. Fuel Rate @(9), lb/kW/hr	---	0.690
(13) Calc. S.S. Gen. Fuel Consumption, lb/hr: (12) x (9)	---	0.690 x 30 = 20.8
(14) Calc. Fuel Consumption For Other Services, lb/hr	---	15.0(heating)
(15) Total Calc. All-Purpose Fuel Consumption lb/hr: (11) + (13) + (14)	---	83.4+20.8+15.0 = 119.2
(16) Calc. All-Purpose Fuel Rate, lb/SHP/hr: (15)/(6) or Heat Balance	1.00	119.2/165 = 0.722
(17) Fuel Rate Correction Factor Based on (7)	1.04	1.04
(18) Specified Fuel Rate, lb/SHP/hr: (16) x (17)	1.00 x 1.04 = 1.04	0.722 x 104 = 0.750
(19) Avg. Endurance Fuel Rate, lb/SHP/hr: (18) x 1.05	1.04 x 1.05 = 1.092	0.750 x 1.05 = 0.787
(20) Endurance Fuel (burnable), tons: (1) x (6) x (19)/(2) x 2240	$\frac{3,000 \times 3,300 \times 1.092}{15 \times 2240}$ = 322	$\frac{1,200 \times 165 \times 0.787}{6 \times 2240}$ = 11.6
(21) Tailpipe Allowance Factor	0.98	0.95
(22) Endurance Fuel Load, tons: (20)/(21)	322/0.98 = 329	11.6/0.95 = 12.2

REFERENCE FOR SOURCE DATA

Design Endurance Power _____
Transmission Efficiency _____
Calc. Prop. Fuel Rate _____
Calc. S.S. Gen. Fuel Rate _____
Calc. Fuel Consumption for PRAIRIE MASKER _____
Calc. Fuel Consumption for Other Services _____
Heat Balance _____
Full Load Displacement _____

DDS 200-1

FIGURE 1. Surface ship endurance calculations.

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